Mathematical Formulas and Equations

Definition 1 (Integral):

The definite integral from a to b of f(x) dx equals F(b) - F(a).

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Formal: integral from 0 to infinity of e^(-x^2) dx = sqrt(pi)/2
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Theorem 1 (Derivative Chain Rule):

If h(x) = f(g(x)), then h'(x) = f'(g(x)) * g'(x).

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Equation: d/dx [f(g(x))] = f'(g(x)) * g'(x)
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Proof:

Apply the definition of the derivative to the composed function. The limit definition gives us h'(x) = lim_{h->0} (f(g(x+h)) - f(g(x))) / h. Factoring yields the result.

Corollary 1:

For f(x) = x^n, the derivative is f'(x) = n * x^(n-1).

Definition 2 (Matrix Multiplication):

Given matrices A of shape m x k and B of shape k x n, their product C = A * B has entry C[i,j] = sum over r of A[i,r] * B[r,j]. This is the fundamental matrix product for linear algebra and neural network computations.

Equation 3:

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C[i,j] = sum_{r=1}^{k} A[i,r] * B[r,j]
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